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Methods of manufacturing semiconductor devices

Abstract

A method of manufacturing a semiconductor device includes forming a light blocking film configured to block first light within a first wavelength band on an edge region of an upper surface of a light-transmitting carrier substrate; forming a photosensitive adhesive layer on the upper surface of the light-transmitting carrier substrate to at least partially cover the light blocking film; bonding a product substrate to the upper surface of the light-transmitting carrier substrate using the photosensitive adhesive layer; partially curing the photosensitive adhesive layer by irradiating the light through the light-transmitting carrier substrate, wherein a portion of the photosensitive adhesive layer overlapping the light blocking film is not cured; processing the product substrate to form a plurality of semiconductor devices after the partially curing of the photosensitive adhesive layer; and cutting the product substrate such that the plurality of semiconductor devices are cut into a plurality of separate individual semiconductor devices.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9716010	12/2016	Gambino et al.	N/A	N/A
9716023	12/2016	Farrens et al.	N/A	N/A
10872802	12/2019	Chang et al.	N/A	N/A
2013/0048224	12/2012	George et al.	N/A	N/A
2014/0318697	12/2013	Tan et al.	N/A	N/A
2019/0109034	12/2018	Fehkuhrer	N/A	N/A
2019/0165037	12/2018	Chae	N/A	H10H 20/835

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2008060361	12/2007	JP	N/A
101392133	12/2013	KR	N/A
101617316	12/2015	KR	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims benefit of priority to Korean Patent Application No. 10-2021-0158391 filed on Nov. 17, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

(2) The present disclosure relates to methods of manufacturing semiconductor devices.

(3) In general, in a process of manufacturing a semiconductor device (e.g., a semiconductor package), a carrier substrate for supporting a product substrate may be used in order to suppress occurrence of warpage of the product substrate. As an adhesive layer between the product substrate and the carrier substrate, an energy-curable resin (e.g., an epoxy based film) such as a thermosetting resin may be used. However, there may be a problem in that a portion fixed by the

adhesive layer during the process could be damaged, or in a detachment process from the product substrate, a complicated subsequent process may be required to remove an adhesive remaining on the product substrate.

SUMMARY

(4) An aspect of the present disclosure is to provide a method of manufacturing a semiconductor device that includes easily detaching a carrier substrate from a product substrate.

(5) According to an aspect of the present disclosure, a method of manufacturing a semiconductor device comprises forming a light blocking film configured to block first light within a first wavelength band on an edge region of an upper surface of a light-transmitting carrier substrate; forming a photosensitive adhesive layer on the upper surface of the light-transmitting carrier substrate to at least partially cover the light blocking film; bonding a product substrate to the upper surface of the light-transmitting carrier substrate using the photosensitive adhesive layer; partially curing the photosensitive adhesive layer by irradiating the first light through the light-transmitting carrier substrate, wherein a portion of the photosensitive adhesive layer overlapping the light blocking film is not cured; processing the product substrate to form a plurality of semiconductor devices, after the partially curing of the photosensitive adhesive layer; and cutting the product substrate such that the plurality of semiconductor devices are cut into a plurality of separate individual semiconductor devices.

(6) According to an aspect of the present disclosure, a method of manufacturing a semiconductor device comprises forming a light blocking film blocking first light having a first wavelength on an edge region of an upper surface of a light-transmitting carrier substrate, the light blocking film having a first width; forming a photosensitive adhesive layer curable by the first light on the upper surface of the light-transmitting carrier substrate to at least partially cover the light blocking film; bonding a product substrate to the upper surface of the light-transmitting carrier substrate using the photosensitive adhesive layer; partially curing the photosensitive adhesive layer by irradiating the first light through the light-transmitting carrier substrate, wherein a portion of the photosensitive adhesive layer overlapping the light blocking film is not cured, and adheres to the product substrate; processing the product substrate to form a plurality of semiconductor devices after the partially curing of the photosensitive adhesive layer, wherein the product substrate has a dummy region surrounding the plurality of semiconductor devices, the dummy region having a second width equal to or wider than the first width; and cutting the plurality of semiconductor devices together with the product substrate, wherein the portion of the photosensitive adhesive layer overlapping the light blocking film is removed in the cutting.

(7) According to an aspect of the present disclosure, a method of manufacturing a semiconductor device comprises forming a light blocking film configured to block first light within a first wavelength band on an edge region of an upper surface of a light-transmitting carrier substrate; forming a photosensitive adhesive layer on the upper surface of the light-transmitting carrier substrate to at least partially cover the light blocking film; bonding a product substrate to the upper surface of the light-transmitting carrier substrate using the photosensitive adhesive layer; partially curing the photosensitive adhesive layer by irradiating the first light through the light-transmitting carrier substrate, wherein a portion of the photosensitive adhesive layer overlapping the light blocking film is not cured, but adheres to the product substrate; processing the product substrate to form a plurality of semiconductor devices after the partially curing of the photosensitive adhesive layer; attaching adhesive tape to an upper surface of the product substrate on which the plurality of semiconductor devices are formed; curing the portion of the photosensitive adhesive layer overlapping the light blocking film by irradiating second light having a second wavelength that is different from the first wavelength through the light blocking film; and detaching the product substrate from the light-transmitting carrier substrate.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The above and other aspects, features, and advantages of the present disclosure will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:

(2) FIG. **1** is a cross-sectional view illustrating a carrier structure according to an embodiment of the present disclosure.

(3) FIG. **2** is a plan view illustrating the carrier structure of FIG. **1**.

(4) FIGS. **3**A and **3**B are cross-sectional views illustrating a portion (a bonding process of an object to be processed) of a method of manufacturing a semiconductor package according to an embodiment of the present disclosure.

(5) FIG. **4** is a plan view illustrating FIG. **3**B, taken along line I-I′.

(6) FIGS. **5**A to **5**E are cross-sectional views illustrating another portion of a method of manufacturing a semiconductor package according to an embodiment of the present disclosure.

(7) FIGS. **6**A and **6**B are cross-sectional views illustrating a cutting process in a method of manufacturing a semiconductor package according to an embodiment of the present disclosure.

(8) FIGS. **7**A to **7**D are plan views illustrating a carrier structure according to various embodiments of the present disclosure.

(9) FIG. **8** is a plan view illustrating a carrier structure according to an embodiment of the present disclosure.

(10) FIGS. **9**A to **9**D are cross-sectional views illustrating a method of manufacturing a semiconductor package according to an embodiment of the present disclosure.

(11) FIGS. **10**A and **10**B are plan views illustrating a carrier structure according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

(12) Hereinafter, an embodiment of the present disclosure will be described in detail with reference to the accompanying drawings.

(13) FIG. **1** is a cross-sectional view illustrating a carrier structure according to an embodiment of the present disclosure, and FIG. **2** is a plan view illustrating the carrier structure of FIG. **1**.

(14) Referring to FIGS. **1** and **2**, a carrier structure **120** according to the present embodiment may include a light-transmitting carrier substrate **121**, and a light blocking film **125** on an edge region **121**T**1** of an upper surface **121**T of the light-transmitting carrier substrate **121**. The upper surface **121**T of the light-transmitting carrier substrate may also have an inner region **121**T**2**. A photosensitive adhesive layer **130** may be on an upper surface of the carrier structure **120** to at least partially cover the light blocking film.

(15) The carrier structure **120** may be used as a temporary support for supporting a product substrate **200**S (indicated by a dotted line) to manufacture a semiconductor device (e.g., a semiconductor package) during a subsequent processing operation. In some embodiments, the product substrate **200**S may be a panel or a wafer for a plurality of semiconductor packages. The carrier structure **120** employed in the present embodiment has been illustrated in a form having a rectangular shape corresponding to a shape of the product substrate **200**S, to support the product substrate **200**S, which may be a rectangular panel, but the present disclosure is not limited thereto. In other embodiments, the carrier structure **120** and may also have a corresponding circular shape for supporting the product substrate **200**S, which may be a wafer (see FIG. **8**).

(16) The photosensitive adhesive layer **130** may include a photosensitive adhesive resin that may be at least partially cured by irradiating light (e.g., ultraviolet light) of a specific wavelength band such that adhesive force maintained during a processing process is lost or weakened. For example, the photosensitive adhesive layer **130** may include photosensitive polyimide (PSPI).

(17) The light-transmitting carrier substrate **121** may be a substrate capable of transmitting light having a wavelength for at least partially curing the photosensitive adhesive layer **130**. For example, the light-transmitting carrier substrate **121** may be a glass substrate. The light blocking film **125** may be a pattern of a material capable of blocking some or all of first light having a first wavelength (e.g., Ultraviolet A (UV-A) or Ultraviolet B (UV-B)). The first wavelength may be within a wavelength band for at least partially curing the photosensitive adhesive layer **130**. For example, the first light may be light in a UV-A band (e.g., 320 nanometers (nm) to 400 nm) or light in a UV-B band (e.g., 280 nm to 320 nm).
(18) The photosensitive adhesive layer **130** may include a first portion **130***a* located on the light blocking film **125**, and a second portion **130***b* located in a region of the upper surface of the light-transmitting carrier substrate **121** surrounded by the light blocking film **125**. It will be understood that "an element A surrounds an element B" (or similar language) as used herein means that the element A is at least partially around the element B but does not necessarily mean that the element A completely encloses the element B. The light blocking film **125** may be configured to block the first light for curing the photosensitive adhesive layer **130**. For example, the light blocking film **125** may be a metal material or a polymer material, and in some embodiments, may include a wavelength selective blocking/transmitting material.
(19) When the first light is irradiated onto a lower surface **121**L of the light-transmitting carrier substrate **121**, the first portion **130***a* of the photosensitive adhesive layer **130** may maintain adhesive force due to blocking of the first light by the light blocking film **125**, and the second portion **130***b* of the photosensitive adhesive layer **130** may be cured by irradiation of the first light to lose or weaken adhesive force.
(20) Since the first portion **130***a* of the photosensitive adhesive layer **130** may be adhered to the edge region **121**T**1** of the product substrate **200**S, the product substrate **200**S may be stably supported on the carrier substrate **121** even with a relatively small contact area during a subsequent processing operation. The product substrate **200**S may have a dummy region surrounding one or more regions of a plurality of semiconductor devices to be manufactured from the product substrate **200**S, and the light blocking film **125** may have a width equal to or narrower than a width of the dummy region (see FIG. **3**B). This dummy region may be provided as a clamping region for handling the product substrate **200**S during a subsequent processing operation. For example, the width of the light blocking film **125** may be in a range of 5 millimeters (mm) to 30 mm in some embodiments.
(21) According to the present embodiment, a portion (e.g., **130***b*) of the photosensitive adhesive layer **130** to be cured which contacts one or more of the regions of the plurality of semiconductor devices (i.e., the semiconductor packages) of the product substrate **200**S may have weak or virtually absent adhesive force. Therefore, damage due to mechanical impact in a subsequent processing operation, contamination due to residual adhesive material after detachment, or the like may be effectively prevented.
(22) In the present embodiment, although the light blocking film **125** is illustrated to be formed of only an edge pattern continuously surrounding the edge region, the light blocking film **125** may further include a pattern extending into the inner region **121**T**2** (please refer to FIGS. **7**A to **7**C) or the edge region, or may include other types of edge patterns non-continuously surrounding the edge region (see FIG. **7**D).
(23) In some embodiments, the light blocking film **125** may be configured to pass second light having a second wavelength therethrough, wherein the second wavelength is different from the first wavelength. The second wavelength may be within a wavelength band (e.g., an ultraviolet band) capable of at least partially curing the photosensitive adhesive layer **130**. When the second light is irradiated onto the lower surface **121**L of the light-transmitting carrier substrate **121** after the subsequent processing operation is completed, the second light may pass through the light-blocking film **125** to be irradiated onto the first portion **130***a* of the photosensitive adhesive layer **130**.

Therefore, the first portion **130***a* may be at least partially cured similarly to the second portion **130***b* to lose or weaken adhesive force. As a result, the product substrate **200**S may be easily detached from the carrier structure **120**.
(24) The photosensitive adhesive layer **130** may be a bonding layer curable by ultraviolet light of UV-A or UV-B band. In an embodiment, the light blocking film **125** may be configured to selectively block first light of the UV-A band (e.g., 320 nm to 400 nm), and may be configured to selectively pass second light of the UV-B band (e.g., 280 nm to 320 nm) therethrough. The light blocking film **125** may include, for example, butylmethoxydibenzoylmethane, dibenzoylmethane, oxybenzene, benzophenone-3, benzophenone-8, mexoryl-SX, phenylbenzimidazole sulfonic acid (PSA), or avobenzone. In another embodiment, the light blocking film **125** may be configured to selectively block first light of the UV-B band (e.g., 280 nm to 320 nm), and may be configured to selectively pass second light of the UV-A band (e.g., 320 nm to 400 nm) therethrough. The light blocking film **125** may include, for example, octocrylene, para-aminobenzoic acid, octyl methoxycinnamate, octyl salicylate, or ethylhexyl triazone.
(25) As such, a detachment process of the product substrate **200**S may be implemented by UV irradiation of the first portion **130***a* using the light blocking film **125** (please refer to FIG. **6**A), but the present disclosure is not limited thereto. The product substrate **200**S may be also implemented by removing the first portion in a cutting process for forming a separate individual semiconductor device (see FIG. **5**E).
(26) Hereinafter, a method of manufacturing a semiconductor device (or a semiconductor package) using the carrier structure illustrated in FIGS. **1** and **2** will be described in detail.
(27) FIGS. **3**A and **3**B are cross-sectional views illustrating a portion (a bonding process of an object to be processed) of a method of manufacturing a semiconductor package according to an embodiment of the present disclosure, and FIGS. **5**A to **5**E are cross-sectional views illustrating another portion of a method of manufacturing a semiconductor package according to an embodiment of the present disclosure. In this case, a manufacturing method according to the present embodiment illustrates a manufacturing method of a so-called “panel level package.” Descriptions of components of the present embodiment may refer to the description of the same or similar components of the carrier structure **120** illustrated in FIGS. **1** and **2**, unless otherwise specifically stated.
(28) Referring to FIG. **3**A, a product substrate **200**S may be bonded to a first carrier structure **120_1** to which a photosensitive adhesive layer **130** is applied.
(29) The product substrate **200**S employed in the present embodiment may include a “panel” for manufacturing a plurality of semiconductor packages. Such a panel may be provided on a first adhesive tape **310**. The product substrate **200**S illustrated in FIG. **3**A may be manufactured through the following processes.
(30) First, a frame **210** having a plurality of cavities **210**H may be disposed on the first adhesive tape **310**. The frame **210** employed in the present embodiment may include insulating members **211***a* and **211***b* having a first surface **210**A and a second surface **210**B, located opposite to each other, and may have a wiring structure connecting the first surface **210**A and the second surface **210**B. The insulating members may include first and second insulating layers **211***a* and **211***b*, and the wiring structure may include three-layered wiring patterns **212***a*, **212***b*, and **212***c*, and wiring vias **213***a* and **213***b* connecting them. For example, the first adhesive tape **310** may be a tape having adhesion, including an epoxy resin. The first adhesive tape **310** may be attached to a lower surface of the first insulating layer **211***a*.
(31) Next, a semiconductor chip **220** may be in each of the cavities **210**H. The semiconductor chip **220** may include a semiconductor body **221** having an active surface such as silicon (Si), germanium (Ge), or gallium arsenide (GaAs), a contact pad **225** formed on the active surface, and a passivation layer **222** on the active surface and exposing the contact pad **225**.
(32) Next, an encapsulant **230** for encapsulating the semiconductor chip **220** may be formed. The

encapsulant **230** may extend to an upper surface of the frame **210**. In the present embodiment, although the encapsulant **230** is illustrated to have an upper surface, substantially flat with a third wiring pattern **212***c*, the encapsulant **230** may be formed to at least partially cover a portion of the third wiring pattern **212***c*.

(33) The product substrate **200**S may have a dummy region **200**E surrounding a plurality of semiconductor package regions **200**. As described above, the dummy region **200**E may be provided as a region for handling the product substrate **200**S in a subsequent processing operation. In addition, the product substrate **200**S may have a margin region **200**M located between the plurality of semiconductor package regions **200**. This margin region **200**M may be provided as a scribe lane for cutting into separate individual packages.

(34) As described in FIGS. **1** and **2**, a light blocking film **125** may be formed on an edge region **121**T**1** of an upper surface **121**T of a light-transmitting carrier substrate **121**, and a photosensitive adhesive layer **130** may be formed on the upper surface **121**T of the light-transmitting carrier substrate **121** to cover the light blocking film **125**.

(35) The product substrate **200**S may be bonded to the photosensitive adhesive layer **130**. The photosensitive adhesive layer **130** may have adhesive force for supporting the product substrate **200**S during a subsequent process. The photosensitive adhesive layer **130** may be formed using various processes such as a spin coating process, a roll coating process, or the like. For example, the photosensitive adhesive layer **130** may include photosensitive polyimide.

(36) Next, referring to FIG. **3**B, a photosensitive adhesive layer **130**′ may be partially cured by irradiating first light having a first wavelength through a lower surface **121**L of a light-transmitting carrier substrate **121**.

(37) The photosensitive adhesive layer **130**′ may include a first portion **130***a* located on a light blocking film **125**, and a second portion **130***b*′ located on an upper surface **121**T of the light-transmitting carrier substrate **121** surrounded by the light blocking film **125**. The light blocking film **125** may be configured to block the first light for curing the photosensitive adhesive layer **130**′. The first light may cure the second portion **130***b*′ of the photosensitive adhesive layer **130**′ through the lower surface **121**L of the light-transmitting carrier substrate **121** to lose or weaken adhesive force.

(38) The first portion **130***a*, a portion of the photosensitive adhesive layer **130**′ overlapping the light blocking film **125**, may not be cured. As used herein, when element A is said to “overlap” or is “overlapping” element B, it may refer to the situation where element A is said to extend over or past, and cover a part of, element B in a given direction. Note that element A may overlap element B in a first direction, but may or may not overlap element B in a second direction. For example, since the first light toward the first portion **130***a* of the photosensitive adhesive layer **130**′ may be blocked by the light blocking film **125**, adhesive force of the uncured first portion **130***a* may be maintained. A width Wa of the light blocking film **125** may be equal to or narrower than a width W**1** of a dummy region **200**E. For example, the width of the light blocking film **125** may be in a range of 5 mm to 30 mm. A width W**2** of a margin region **200**M may be narrower than the width W**1** of the dummy region **200**E.

(39) FIG. **4** is a plan view illustrating FIG. **3**B, taken along line I-I′. Descriptions of components of the present embodiment may refer to the description of the same or similar components of the carrier structure **120** illustrated in previous Figures, unless otherwise specifically stated.

(40) Referring to FIG. **4**, the first portion **130***a* overlapping the light blocking film **125** may maintain adhesive force in an uncured state, and the second portion **130***b*′ that does not overlap the light blocking film **125** may be cured to lose or weaken adhesive force. Although the first portion **130***a* has a relatively small area, the first portion **130***a* may surround an edge region. Therefore, the product substrate **200**S may be stably maintained on a first carrier structure **120_1** in a subsequent processing operation.

(41) Next, after the partial curing is performed, a processing operation of the product substrate

200S for forming the plurality of semiconductor package regions **200** may be performed (please refer to FIGS. **5**A to **5**D). Descriptions of components of the present embodiment may refer to the description of the same or similar components of the carrier structure **120** illustrated in previous Figures, unless otherwise specifically stated.

(42) First, referring to FIG. **5**A, after a first adhesive tape **310** is removed from a product substrate **200**S, a first redistribution structure **240** may be formed on the removed surface of the product substrate **200**S.

(43) The formation of the first redistribution structure **240** may be performed by forming a first insulating film **241** using a lamination process or a coating process, forming a via hole in the first insulating film **241**, and forming a first redistribution layer **245**, e.g., a first redistribution pattern **242** and a first redistribution via **243**, using an electroplating process or an electroless plating process. When a photo-imageable dielectric (PID) material is used as the first insulating film **241**, via-holes may be formed as a fine pitch using a photolithography process. The first redistribution structure **240** may include a plurality of first insulating films **241** and a plurality of first redistribution layers **245** respectively formed on the plurality of first insulating films **241**.

(44) Additionally, a passivation layer **260** and an underbump metal layer (UBM layer) **270** may be formed. The passivation layer **260** may be formed on a lower surface of the first redistribution structure **240**, a plurality of openings exposing a portion of the first redistribution layer **245** may be formed in the passivation layer **260**, and the underbump metal layer **270** may be formed on the passivation layer **260** to be connected to a region of the first redistribution layer **245** through the plurality of openings.

(45) Next, referring to FIG. **5**B, a second carrier structure **120_2** may be bonded to the product substrate **200**S.

(46) This process may be introduced when a second redistribution structure (see, e.g., **250** of FIG. **5**D), which may be a backside redistribution layer, is additionally formed. Similar to the first carrier structure **120_1** of FIG. **3**A described above, a light blocking film **125** may be formed on an edge region **121**T**1** of an upper surface **121**T of a light-transmitting carrier substrate **121**, and a photosensitive adhesive layer **130′** may be formed on the upper surface **121**T of the light-transmitting carrier substrate **121** to cover the light blocking film **125**. Next, the second carrier structure **120_2** may be bonded to an upper surface of the product substrate **200**S, and a surface on which the first redistribution structure **240** is formed on the photosensitive adhesive layer **130′**. Next, a second portion **130***b*′ of the photosensitive adhesive layer **130′** may be at least partially cured by irradiating the first light through the light-transmitting carrier substrate **121**, and the second carrier structure **120_2** and the product substrate **200**S may be bonded by a first portion **130***a* overlapping the light blocking film **125**.

(47) Next, referring to FIG. **5**C, the product substrate **200**S bonded to the second carrier structure **120_2** may be detached from the first carrier structure **120_1**.

(48) The detachment of the first carrier structure **120_1** may be performed by irradiating second light having a second wavelength that is different from the first wavelength through the light blocking film **125** to at least partially cure a portion of the photosensitive adhesive layer **130″** overlapping the light blocking film **125**, for example, the first portion **130***a*′.

(49) The photosensitive adhesive layer **130** may be a bonding layer that may be curable in the ultraviolet band.

(50) In some embodiments, the first light may be light of the UV-A band, and the second light may be light of the UV-B band. For example, the light blocking film **125** may include butylmethoxydibenzoylmethane, dibenzoylmethane, oxybenzene, benzophenone-3, benzophenone-8, mexoryl SX, phenylbenzimidazole sulfonic acid, or avobenzone. In another embodiment, the first light may be light of the UV-B band, and the second light may be light of the UV-A band. For example, the light blocking film **125** may include octocrylene, para-aminobenzoic acid, octylmethoxycinnamate, octylsalicylate, or ethylhexyltriazone.

(51) Next, referring to FIG. **5**D, a second redistribution structure **250** may be formed on a surface from which the first carrier structure **120_1** is removed.
(52) Formation of the second redistribution structure **250** may include forming a second insulating film **251**, forming a via hole in the second insulating film **251**, and forming a second redistribution layer **255**, e.g., a second redistribution pattern **252** and a second redistribution via **253**. The second insulating film **251** may include a PID material.
(53) As described above, a portion (e.g., **130***b*′) of the photosensitive adhesive layer **130** to be cured which contacts a plurality of semiconductor package regions **200** of the product substrate **200**S may have weak or virtually absent adhesive force. Therefore, even though mechanical impact is generated in the formation of the second redistribution structure **250**, damage to the first redistribution structure **240** (in particular, the UBM layer **270**) previously formed in each of the plurality of semiconductor package regions **200** may be prevented.
(54) Next, referring to FIG. **5**E, the product substrate **200**S illustrated in FIG. **5**D may be cut into a plurality of separate individual semiconductor devices **200**.
(55) In this cutting process, a portion (e.g., **130***a*) of the photosensitive adhesive layer **130** overlapping the light blocking film **125** may be partially removed. Since the previously cured second portion **130***b*′ has previously lost adhesive force, an individualized semiconductor package **200** may be detached from the first portion **130***a* still having adhesive force by the partial removal of this process.
(56) A process of additionally separating using irradiation of another ultraviolet light may be performed similarly to the process of FIG. **5**C, without detaching from the second carrier structure by the cutting process. FIGS. **6**A and **6**B are cross-sectional views illustrating a cutting process in a method of manufacturing a semiconductor package according to an embodiment of the present disclosure. Descriptions of components of the present embodiment may refer to the description of the same or similar components of the carrier structure **120** illustrated in previous figures, unless otherwise specifically stated.
(57) Referring to FIG. **6**A, a second adhesive tape **320** may be attached to an upper surface of a product substrate **200**S on which a plurality of semiconductor packages **200** are formed, and a first portion **130***a*′ of a photosensitive adhesive layer **130**′ may be at least partially cured by irradiating second light within a second wavelength that is different from the first wavelength through a light blocking film **125**. Since adhesive force of a previously cured second portion **130***b*′ is lost, the photosensitive adhesive layer **130**′ and the product substrate **200**S may be detached by additional curing of the first portion **130***a*′.
(58) Next, referring to FIG. **6**B, after detaching the product substrate **200**S from a light-transmitting carrier substrate **121**, the product substrate **200**S may be cut into the plurality of semiconductor packages **200**.
(59) As described above, a portion (e.g., **130***b*′) of the photosensitive adhesive layer **130**′ to be cured which contacts the plurality of semiconductor packages **200** of the product substrate **200**S may have weak or virtually absent adhesive force. Therefore, damage or contamination of the first redistribution structure **240** (in particular, the UBM layer **270**) previously formed in each of the semiconductor package regions **200** may be effectively prevented.
(60) FIGS. **7**A to **7**D are plan views illustrating a carrier structure according to various embodiments of the present disclosure.
(61) Referring to FIG. **7**A, it can be understood that a carrier structure **120**A according to the present embodiment is similar to that illustrated in FIGS. **1** and **2**, except that a light blocking film **125** has an inner pattern **125***b* extending to an inner region **121**T**2** in addition to an edge pattern **125***a*, as illustrated in FIGS. **1** and **2**. Descriptions of components of the present embodiment may refer to the description of the same or similar components of the carrier structure **120** illustrated in FIGS. **1** and **2**, unless otherwise specifically stated.
(62) The light blocking film **125** employed in the present embodiment may be formed on an upper

surface **121**T of a light-transmitting carrier substrate **121** having a rectangular structure similar to that of the previous embodiments, and may include an edge pattern **125***a* formed along an edge region **121**T**1** of the upper surface **121**T, and an inner pattern **125***b* extending to the inner region **121**T**2** surrounded by the edge region. The inner pattern **125***b* may be configured to connect two opposite sides to intersect the inner region **121**T**2** of the upper surface.
(63) The light blocking film **125** may have a width equal to or narrower than the width W**1** of the dummy region (**200**E of FIG. **3**B). For example, the width of the light blocking film **125** may be in a range of 5 mm to 30 mm. In some embodiments, the width of the light blocking film **125** may range of 10 mm to 25 mm. The inner pattern **125***b* may have a width different from a width of the edge pattern **125***a*.
(64) Referring to FIG. **7**B, it can be understood that a carrier structure **120**B according to the present embodiment is similar to that illustrated in FIGS. **1** and **2**, except that a light blocking film **125** has first and second inner patterns **125***b***1** and **125***b***2** extending and intersecting in different directions (e.g., in a vertical direction) in an inner region, other than an edge pattern **125***a*. Descriptions of components of the present embodiment may refer to the description of the same or similar components of the carrier structure **120** illustrated in FIGS. **1** and **2**, unless otherwise specifically stated.
(65) The light blocking film **125** employed in the present embodiment may include an edge pattern **125***a* formed along an edge region of an upper surface **121**T, a first inner pattern **125***b***1** extending in a first direction, and a second inner pattern **125***b***2** extending in a second direction perpendicular to the first direction. Each of the first and second inner patterns **125***b***1** and **125***b***2** may have a width Wb that is narrower than a width Wa of the edge pattern **125***a*. As illustrated in FIG. **3**B, a product substrate **200**S to be bonded may include a dummy region **200**E surrounding a plurality of semiconductor package regions **200**, and a margin region **200**M located between the plurality of semiconductor packages **200**.
(66) The edge pattern **125***a* may have a width equal to or narrower than the width W**1** of the dummy region **200**E. For example, the width of the edge pattern **125***a* may be in a range of 5 mm to 30 mm. In some embodiments, the width of the edge pattern **125***a* may range of 10 mm to 25 mm. The inner pattern **125***b* may have a width Wb that is narrower than a width Wa of the edge pattern **125***a*. Each of the first and second inner patterns **125***b***1** and **125***b***2** may have a width that is equal to or narrower than a width of the margin region **200**M. Also, the first and second inner patterns **125***b***1** and **125***b***2** may be arranged to overlap the margin region **200**M, respectively.
(67) Referring to FIG. **7**C, it can be understood that a carrier structure **120**C according to the present embodiment is similar to that illustrated in FIGS. **1** and **2**, except that a light blocking film **125** has first and second inner patterns **125***b***1** and **125***b***2** extending and intersecting in different directions (e.g., in a vertical direction) in an inner region, other than an edge pattern **125***a*, and the second inner patterns **125***b***2** may be arranged as a plurality of second inner patterns **125***b***2**. Descriptions of components of the present embodiment may refer to the description of the same or similar components of the carrier structure **120** illustrated in FIGS. **1** and **2**, unless otherwise specifically stated.
(68) The light blocking film **125** employed in the present embodiment, similarly to the embodiment illustrated in FIG. **7**B, may include an edge pattern **125***a* formed along an edge region of an upper surface **121**T, a first inner pattern **125***b***1** extending in a first direction, and a second inner pattern **125***b***2** extending in a second direction that is perpendicular to the first direction. The second inner patterns **125***b***2** may be arranged as a plurality of second inner patterns **125***b***2**. The first inner pattern **125***b***1** as well as the plurality of second inner patterns **125***b***2** may be arranged to respectively overlap a margin region **200**M. The first and second inner patterns **125***b***1** and **125***b***2** may have a width that is equal to or narrower than a width of the margin region **200**M. In addition, each of the first and second inner patterns **125***b***1** and **125***b***2** may have a width that is narrower than a width Wa of the edge pattern **125***a*, and may have different widths Wb**1** and Wb**2**, as in the

present embodiment.

(69) In the present embodiment, although an example in which the second inner pattern **125*b*2** is provided in plural is illustrated, the first inner pattern **125*b*1** may be provided in plural and the two inner patterns **125*b*1** and **125*b*2** may be provided in plural.

(70) Referring to FIG. **7**D, it can be understood that a carrier structure **120**D according to the present embodiment is similar to that illustrated in FIGS. **1** and **2**, except that a light blocking film **125′** has a discontinuous pattern. Descriptions of components of the present embodiment may refer to the description of the same or similar components of the carrier structure **120** illustrated in FIGS. **1** and **2**, unless otherwise specifically stated.

(71) Unlike the embodiments of FIGS. **1** and **2**, the light blocking film **125′** employed in the present embodiment may. Such a discontinuous pattern may be similarly applied to the inner patterns **125*b***, **125*b*1**, and **125*b*2** employed in the previous embodiments.

(72) The carrier structure **120** according to the previous embodiments is illustrated to form a rectangular shape corresponding to a shape of the product substrate **200**S as a rectangular panel, but the present disclosure is not limited thereto, and may have a corresponding circular shape for supporting the product substrate, which may be a wafer.

(73) FIG. **8** is a plan view illustrating a carrier structure according to an embodiment of the present disclosure.

(74) Referring to FIG. **8**, it can be understood that a carrier structure **120**E according to the present embodiment is similar to that illustrated in FIGS. **1** and **2**, except that a light-transmitting carrier substrate **121** on which a light blocking film **125** is formed has a circular upper surface **121**T. Descriptions of components of the present embodiment may refer to the description of the same or similar components of the carrier structure **120** illustrated in FIGS. **1** and **2**, unless otherwise specifically stated.

(75) The light-transmitting carrier substrate **121** employed in the present embodiment may have a circular planar shape corresponding to a wafer, which may be a product substrate. A product substrate on which the carrier structure **120**E according to the present embodiment is used may be a wafer on which a plurality of semiconductor devices for a so-called “wafer level package” are implemented. The light blocking film **125** may be continuously formed along an edge region of the upper surface **121**T of the light-transmitting carrier substrate **121** having a circular shape. The light blocking film **125** is not limited to a continuous edge pattern, and may have other patterns similar to the various patterns illustrated in the previous embodiments (FIGS. **7**A to **7**B).

(76) FIGS. **9**A to **9**D are cross-sectional views illustrating a method of manufacturing a semiconductor package according to an embodiment of the present disclosure. Descriptions of components of the present embodiment may refer to the description of the same or similar components of the carrier structure **120** illustrated in previous Figures, unless otherwise specifically stated.

(77) Referring to FIG. **9**A, a semiconductor wafer **400**W may be bonded to a carrier structure **120**E to which a photosensitive adhesive layer **130** is applied.

(78) The semiconductor wafer **400**W employed in the present embodiment may be a “semiconductor wafer” on which a plurality of first semiconductor chips **400**A may be implemented. In the present embodiment, each of the plurality of first semiconductor chips **400**A may have a structure provided as a base substrate of a semiconductor package. Each of the plurality of first semiconductor chips **400**A may include a semiconductor substrate **410** having an active surface and a non-active surface located opposite thereto, a first pad **420** on the active surface of the semiconductor substrate **410**, a second pad **430** on the non-active surface of the substrate **410**, and a through-electrode **450** passing through the semiconductor substrate **410** and connecting the first and second pads **420** and **430**. A connection bump **460** such as a solder ball may be formed on the first pad **420** of the first semiconductor chip **400**A. In the present embodiment, although the first semiconductor chips **400**A is illustrated to be introduced as the base substrates, the first

semiconductor chips **400**A may be implemented as an interposer having a wiring structure.

(79) As illustrated in FIG. **9**A, the semiconductor wafer **400**W on which the connection bump **460** is formed may be attached to the photosensitive adhesive layer **130** on the carrier structure **120**E. The semiconductor wafer **400**W may be attached to the photosensitive adhesive layer **130** such that the connection bump **460** faces the carrier structure **120**E. The connection bump **460** may be covered by the photosensitive adhesive layer **130**. Among lower surfaces of the semiconductor wafer **400**W, a lower surface on which the connection bump **460** is not formed may be bonded to the photosensitive adhesive layer **130**.

(80) As described in the previous embodiment (FIG. **3**A), a light blocking film **125** may be formed on an edge region **121**T**1** of an upper surface **121**T of a light-transmitting carrier substrate **121**, and a photosensitive adhesive layer **130** may be formed on the upper surface **121**T of the light-transmitting carrier substrate **121** to at least partially cover the light blocking film **125**. The semiconductor wafer **400**W may be bonded to the photosensitive adhesive layer **130**.

(81) Next, referring to FIG. **9**B, the photosensitive adhesive layer **130** may be partially cured by irradiating light within a first wavelength band through a lower surface of the light-transmitting carrier substrate **121**.

(82) The photosensitive adhesive layer **130** may include a first portion **130***a* located on the light blocking film **125**, and a second portion **130***b* located on a region of the upper surface of the light-transmitting carrier substrate **121** surrounded by the light blocking film **125**. The light blocking film **125** may be configured to block the first light for curing the photosensitive adhesive layer **130**. The first light may at least partially cure the second portion **130***b'* of the photosensitive adhesive layer **130** through the lower surface of the light-transmitting carrier substrate **121** to lose or weaken adhesive force.

(83) The first portion **130***a*, which may be a portion of the photosensitive adhesive layer **130** overlapping the light blocking film **125**, may not be cured. For example, since the first light toward the first portion **130***a* of the photosensitive adhesive layer **130** may be blocked by the light blocking film **125**, adhesive force of the uncured first portion **130***a* may be maintained. A width of the light blocking film **125** may be equal to or narrower than a width of a dummy region **400**E. The semiconductor wafer **400**W may have a margin region **400**M located between the plurality of first semiconductor chips **400**A. In the cutting process for singulation, the margin region **400**M may be removed. A width of a margin region **200**M may be narrower than a width of the dummy region **400**E.

(84) Referring to FIG. **9**C, one or more second semiconductor chips **400**B may be stacked on the semiconductor wafer **400**W.

(85) The second semiconductor chips **400**B may be obtained by cutting a second semiconductor wafer on which a plurality of semiconductor chips are manufactured similarly to the first semiconductor wafer **400**W. In the present embodiment, the second semiconductor wafer may be a semiconductor wafer including individual devices of the same type as that of the first semiconductor wafer **400**W. For example, the plurality of second semiconductor chips **400**B may include a second semiconductor substrate **410**, first and second pads **420** and **430**, a through-electrode **450**, and a connection bump **460**. Each of the plurality of second semiconductor chips **400**B may be stacked on the first semiconductor chip **400**A of the first semiconductor wafer **400**W by using a non-conductive film **480**. The first pad **420** of each of the plurality of second semiconductor chips **400**B may be connected to the second pad **430** of the first semiconductor chips by the connection bump **460**.

(86) Referring to FIG. **9**D, similarly to the previous stacking process on the second semiconductor chips **400**B, one or more third semiconductor chips **400**C and one or more fourth semiconductor chips **400**D may be sequentially stacked, respectively, on the semiconductor wafer **400**W. A molding portion **490** may be formed to cover the second to fourth semiconductor chips **400**B, **400**C, and **400**D. A stacking process of the second to fourth semiconductor chips **400**B, **400**C, and

400D may be performed by a reflow process or a thermal compression process. The molding portion **490** may include, for example, an epoxy mold compound (EMC). In a cutting process for singulation, the semiconductor wafer **400**W together with the molding portion **490** may be cut into a semiconductor package **400**.

(87) In this cutting process, a portion (e.g., **130***a*) of the photosensitive adhesive layer **130** overlapping the light blocking film **125** may be partially removed. Since the previously cured second portion **130***b*′ has previously lost adhesive force, the individualized semiconductor package **200** may be detached from the first portion **130***a* having adhesive force by the partial removal of the present process.

(88) The first to fourth semiconductor chips **400**A, **400**B, **400**C, and **400**D employed in the present embodiment may be a memory chip or a logic chip. In some embodiments, the first to fourth semiconductor chips **400**A, **400**B, **400**C, and **400**D may all be the same type of memory chip, and in other embodiments, the first to fourth semiconductor chips **400**A, **400**B, **400**C, and **400**D may be a memory chip, and some may be a logic chip.

(89) For example, the memory chip may be a volatile memory chip such as a dynamic random access memory (DRAM) or a static random access memory (SRAM), or may be a non-volatile memory chip such as a phase-change random access memory (PRAM), a magnetoresistive random access memory (MRAM), a ferroelectric random access memory (FeRAM) or a resistive random access memory (RRAM). Also, the logic chip may be, for example, a microprocessor, an analog device, or a digital signal processor. In certain embodiments, the first to fourth semiconductor chips **400**A, **400**B, **400**C, and **400**D may be a DRAM, and the semiconductor package **200** may be a high bandwidth memory (HBM).

(90) FIGS. **10**A and **10**B are plan views illustrating a carrier structure according to various embodiments of the present disclosure.

(91) Referring to FIG. **10**A, it can be understood that a carrier structure **120**F according to the present embodiment is similar to that illustrated in FIG. **8**, except that a light blocking film **125** has first and second inner patterns **125***b***1** and **125***b***2** extending and intersecting in different directions (e.g., in a vertical direction) in an inner region, other than an edge pattern **125***a*. Descriptions of components of the present embodiment may refer to the description of the same or similar components of the carrier structure **120**E illustrated in FIG. **8**, unless otherwise specifically stated.

(92) The light blocking film **125** employed in the present embodiment may include an edge pattern **125***a* formed along an edge region of an upper surface **121**T, a first inner pattern **125***b***1** extending in a first direction, and a second inner pattern **125***b***2** extending in a second direction perpendicular to the first direction. Each of the first and second inner patterns **125***b***1** and **125***b***2** may have a width Wb that is narrower than a width Wa of the edge pattern **125***a*. Each of the first and second inner patterns **125***b***1** and **125***b***2** may have a width Wb that is equal to or narrower than a width of a margin region **200**M. Also, the first and second inner patterns **125***b***1** and **125***b***2** may be arranged to overlap the margin region **200**M, respectively.

(93) Referring to FIG. **10**B, it can be understood that a carrier structure **120**G according to the present embodiment is similar to that illustrated in FIG. **8**, except that a light blocking film **125**′ has a discontinuous pattern. Descriptions of components of the present embodiment may refer to the description of the same or similar components of the carrier structure **120**E illustrated in FIG. **8**, unless otherwise specifically stated.

(94) Unlike the embodiment of FIG. **8**, the light blocking film **125**′ employed in the present embodiment may include a plurality of edge patterns discontinuously formed along an edge region. Such a discontinuous pattern may be similarly applied to the inner patterns **125***b***1** and **125***b***2** employed in the previous embodiments.

(95) According to an embodiment of the present disclosure, damage to a product substrate may be minimized and the product substrate may be easily detached from a photosensitive adhesive layer, by partially curing the photosensitive adhesive layer bonded to the product substrate using a light

blocking film, in a process of manufacturing a semiconductor device.
(96) Various advantages and effects of the present disclosure are not limited to the above, and will be more easily understood in the process of describing specific embodiments of the present disclosure.
(97) While example embodiments have been illustrated and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present disclosure as defined by the appended claims.

Claims

1. A method of manufacturing a semiconductor device, comprising: forming a light blocking film configured to block first light within a first wavelength band on an edge region of an upper surface of a light-transmitting carrier substrate; forming a photosensitive adhesive layer on the upper surface of the light-transmitting carrier substrate to at least partially cover the light blocking film; bonding a product substrate to the upper surface of the light-transmitting carrier substrate using the photosensitive adhesive layer; partially curing the photosensitive adhesive layer by irradiating the first light through the light-transmitting carrier substrate, wherein a portion of the photosensitive adhesive layer overlapping the light blocking film is not cured; processing the product substrate to form a plurality of semiconductor devices, after the partially curing of the photosensitive adhesive layer; and cutting the product substrate such that the plurality of semiconductor devices are cut into a plurality of separate individual semiconductor devices.
2. The method of claim 1, wherein the light blocking film comprises a pattern continuously formed along the edge region.
3. The method of claim 1, wherein the product substrate has a dummy region surrounding the plurality of semiconductor devices, and wherein the light blocking film has a width equal to or narrower than a width of the dummy region.
4. The method of claim 3, wherein the width of the light blocking film is 5 mm to 30 mm.
5. The method of claim 1, further comprising forming the light blocking film to have an edge pattern located in the edge region and an inner pattern extending from the upper surface to an inner region surrounded by the edge region.
6. The method of claim 5, wherein the product substrate has a margin region located between the plurality of semiconductor devices, wherein the inner pattern has a width equal to or narrower than a width of the margin region, and is located to overlap the margin region, and wherein the inner pattern has a width smaller than a width of the edge pattern.
7. The method of claim 1, wherein forming the light blocking film comprises forming the light blocking film to have one or more edge patterns discontinuously formed along the edge region.
8. The method of claim 5, wherein the inner pattern comprises at least one first inner pattern extending in a first direction, and at least one second inner pattern extending in a second direction intersecting the first direction.
9. The method of claim 1, further comprising: before the cutting of the product substrate, attaching adhesive tape to an upper surface of the product substrate on which the plurality of semiconductor devices are formed; curing the portion of the photosensitive adhesive layer overlapping the light blocking film by irradiating second light having a second wavelength that is outside the first wavelength band through the light blocking film; and detaching the product substrate from the light-transmitting carrier substrate.
10. The method of claim 9, wherein the first light is ultraviolet light of a Ultraviolet A (UV-A) band, and the second light is ultraviolet light of a Ultraviolet B (UV-B) band.
11. The method of claim 10, wherein the light blocking film comprises a material selected from the group consisting of butylmethoxydibenzoylmethane, dibenzoylmethane, oxybenzene, benzophenone-3, benzophenone-8, Mexoryl-SX, phenylbenzimidazole sulfonic acid (PSA), and/or

Avobenzone.

12. The method of claim 9, wherein the first light is ultraviolet light of a Ultraviolet B (UV-B) band, and the second light is ultraviolet light of a Ultraviolet A (UV-A) band.

13. The method of claim 12, wherein the light blocking film comprises a material selected from the group consisting of octocrylene, para-aminobenzoic acid, octyl methoxycinnamate, octyl salicylate, and/or ethylhexyltriazone.

14. The method of claim 1, further comprising removing the portion of the photosensitive adhesive layer overlapping the light blocking film after the processing of the product substrate.

15. A method of manufacturing a semiconductor device, comprising: forming a light blocking film configured to block first light within a first wavelength band on an edge region of an upper surface of a light-transmitting carrier substrate, the light blocking film having a first width; forming a photosensitive adhesive layer curable by the first light on the upper surface of the light-transmitting carrier substrate to at least partially cover the light blocking film; bonding a product substrate to the upper surface of the light-transmitting carrier substrate using the photosensitive adhesive layer; partially curing the photosensitive adhesive layer by irradiating the first light through the light-transmitting carrier substrate, wherein a portion of the photosensitive adhesive layer overlapping the light blocking film is not cured and adheres to the product substrate; processing the product substrate to form a plurality of semiconductor devices, after the partially curing of the photosensitive adhesive layer, wherein the product substrate has a dummy region surrounding the plurality of semiconductor devices, the dummy region having a second width that is equal to or wider than the first width; and cutting the plurality of semiconductor devices together with the product substrate, wherein the portion of the photosensitive adhesive layer overlapping the light blocking film is removed by the cutting.

16. The method of claim 15, wherein the product substrate comprises a frame having a first surface and a second surface opposite to each other, and having a plurality of through-holes, a plurality of semiconductor chips respectively in the plurality of through-holes, an encapsulant sealing the plurality of semiconductor chips, and a first redistribution structure on the first surface of the frame and having a first redistribution layer connected to one or more of the plurality of semiconductor chips, and wherein the bonding of the product substrate comprises bonding the product substrate such that the first redistribution structure faces the photosensitive adhesive layer.

17. The method of claim 16, wherein the product substrate further comprises a passivation layer on the first redistribution structure and having an opening and an underbump metal layer in the opening of the passivation layer and connected to the first redistribution layer, wherein the bonding of the product substrate comprises bonding the product substrate such that the passivation layer faces the photosensitive adhesive layer.

18. The method of claim 16, wherein the frame has a wiring structure connecting the first surface and the second surface, and wherein the processing of the product substrate comprises forming a second redistribution structure having a second redistribution layer connected to the wiring structure on the second surface of the frame.

19. A method of manufacturing a semiconductor device, comprising: forming a light blocking film configured to block first light within a first wavelength band on an edge region of an upper surface of a light-transmitting carrier substrate; forming a photosensitive adhesive layer on the upper surface of the light-transmitting carrier substrate to at least partially cover the light blocking film; bonding a product substrate to the upper surface of the light-transmitting carrier substrate using the photosensitive adhesive layer; partially curing the photosensitive adhesive layer by irradiating the first light through the light-transmitting carrier substrate, wherein a portion of the photosensitive adhesive layer overlapping the light blocking film is not cured, but adheres to the product substrate; processing the product substrate to form a plurality of semiconductor devices, after the partially curing of the photosensitive adhesive layer; attaching adhesive tape to an upper surface of the product substrate on which the plurality of semiconductor devices are formed; curing the portion of

the photosensitive adhesive layer overlapping the light blocking film by irradiating second light having a second wavelength that is outside the first wavelength band through the light blocking film; and detaching the product substrate from the light-transmitting carrier substrate.

20. The method of claim 19, further comprising cutting the plurality of semiconductor devices together with the product substrate, after the detaching of the product substrate.